

1. Which type of machine learning is used when the output variable is continuous?
 - a) Classification
 - b) Clustering
 - c) Regression
 - d) Reinforcement Learning

Answer: Regression

2. In which scenario would you use classification instead of regression?
 - a) Predicting house prices based on area
 - b) Predicting whether an email is spam or not
 - c) Predicting the temperature of a city
 - d) Predicting the total sales of a store

Answer: b) Predicting whether an email is spam or not

3. Which of the following is an unsupervised learning algorithm?
 - a) Decision Tree
 - b) K-Means
 - c) Linear Regression
 - d) Logistic Regression

Answer: b) K-Means

4. What does the term "overfitting" refer to in machine learning?
 - a) When the model does not learn from the data
 - b) When the model fits the training data too well but fails on new data
 - c) When the model generalizes well to unseen data
 - d) When the model cannot capture complex patterns

Answer: b) When the model fits the training data too well but fails on new data

5. Which evaluation metric is most commonly used for classification problems?
 - a) Mean Squared Error (MSE)
 - b) R-squared
 - c) Accuracy
 - d) Root Mean Squared Error (RMSE)

Answer: c) Accuracy

6. What is the main assumption of linear regression?
 - a) The relationship between dependent and independent variables is linear

- b) The data is always normally distributed
- c) There is no correlation between independent variables
- d) It can only be used for categorical data

Answer: a) The relationship between dependent and independent variables is linear

7. Which clustering algorithm is best for detecting outliers?
- a) K-Means
 - b) DBSCAN
 - c) Hierarchical Clustering
 - d) Linear Regression

Answer: b) DBSCAN

8. Which regularization technique adds both L1 and L2 penalties to a regression model?
- a) Ridge Regression
 - b) Lasso Regression
 - c) Elastic Net
 - d) Polynomial Regression

Answer: c) Elastic Net

9. What is the purpose of the kernel trick in Support Vector Machines (SVM)?
- a) To reduce training time
 - b) To map data into a higher-dimensional space for better classification
 - c) To normalize data
 - d) To make the model interpret results more easily

Answer: b) To map data into a higher-dimensional space for better classification

10. Which clustering algorithm does not require specifying the number of clusters in advance?
- a) K-Means
 - b) DBSCAN
 - c) KNN
 - d) Decision Tree

Answer: b) DBSCAN

11. What type of function does logistic regression use to convert values into probabilities?
- a) Linear function
 - b) Sigmoid function
 - c) Exponential function
 - d) Softmax function

Answer: Sigmoid function

12. Which metric is commonly used to evaluate a clustering algorithm?

- a) F1-score
- b) Sum of Squared Errors (SSE)
- c) Mean Squared Error
- d) Log-Loss

Answer: b) Sum of Squared Errors (SSE)

13. If your classification model has a high recall but low precision, what does it mean?

- a) The model is predicting too many false positives
- b) The model is predicting too many false negatives
- c) The model has low accuracy
- d) The model is not generalizing well

Answer: a) The model is predicting too many false positives

14. Which technique can be used to reduce the dimensionality of a dataset before applying clustering?

- a) Support Vector Machine
- b) Principal Component Analysis (PCA)
- c) Logistic Regression
- d) Random Forest

Answer: b) Principal Component Analysis (PCA)

15. What is the main limitation of the K-Means clustering algorithm?

- a) It does not scale well for large datasets
- b) It requires the number of clusters to be specified in advance
- c) It cannot handle numerical data
- d) It always produces overlapping clusters

Answer: b) It requires the number of clusters to be specified in advance